Comparative Analysis of Bias Mitigation Techniques in Supervised Machine Learning

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Abstract - Automated decision-making systems are now a big part of important areas like hiring, credit allocation, clinical triage, and criminal risk assessment. However, the results they give often reinforce the unfairness that was built into their training data. This paper offers a methodical, direct comparison of six bias mitigation strategies sourced from three separate intervention stages in the supervised learning pipeline: pre-processing, in-processing, and post-processing. We use three classifiers with different levels of structural complexity—Logistic Regression, Decision Tree, and Random Forest—to test two well-known benchmark datasets (Adult Income and COMPAS). This makes sure that the results are not just the result of one type of algorithm. Fairness is assessed using standard accuracy and macro-F1 measures, as well as Demographic Parity Difference (DPD) and Equal Opportunity Difference (EOD). The findings demonstrate that all interventions considerably reduce group-level difference, although the magnitude of the accuracy-fairness trade-off changes substantially between stages. While in-processing methods achieve the greatest bias reductions at a higher prediction cost, pre-processing techniques yield small but constant improvements in fairness with minimal performance decrease. Because they allow you to alter the result without retraining, post-processing techniques are a helpful middle ground. These findings provide organized evidence for selecting mitigation solutions that satisfy specific accuracy and fairness standards, and they have practical consequences for practitioners who must deal with deployment limits in the real world.

Keywords - Fairness in machine learning, reducing bias, supervised learning, ethical AI, fairness metrics, demographic parity, equal opportunity, and algorithmic accountability.

I. INTRODUCTION

Machine learning systems are being used more and more in important areas, but we can't keep up with how quickly we can check what those systems actually learn. Hiring platforms score résumés, lending algorithms figure out if someone is creditworthy, clinical tools figure out how risky a patient is, and recidivism prediction tools help decide whether to grant bail—all at a speed and scale that no human reviewer can match. But being fast doesn't mean being fair. When a model learns from decades of historical records that show unequal treatment, it doesn't just describe the past; it also makes it a rule, projects it into the future, and uses it in industrial throughput. Outcomes that used to depend on intentional human bias can now happen automatically, thanks to weights and thresholds that are hidden from the people who are most affected by them.

There are many ways that bias can get into these systems, and they often build on each other. Models are not well-calibrated for certain demographic groups when training corpora don't include enough of them. Proxy variables—characteristics that are statistically related to protected traits without naming them directly—let models recreate sensitive differences even when those traits are purposefully left out. Label noise, which comes from biased human judgments in the past, spreads mistakes into future predictions. The outcome is a category of systems that can achieve high scores on overall accuracy metrics while generating consistently unfavourable results for specific subgroups [1].

In response, researchers have suggested interventions at every step of the modelling process. Before a model ever sees the training data, pre-processing methods change it by changing the distribution of examples or their labels to fix structural imbalances. In-processing methods put fairness goals right into the learning algorithm, which limits or punishes the model as it trains. Post-processing techniques work after training is done and adjust predicted scores or decision thresholds. Every family of methods has its own set of assumptions, costs, and practical trade-offs. However, there aren't many comparative studies that keep all other factors the same across all three families.

That gap is directly addressed in this study. This work's principal contributions are:

(1) A single experimental benchmark that compares six representative bias mitigation techniques, two for each pipeline stage, assessed under the same circumstances using the same datasets, classifiers, preprocessing procedures, and evaluation criteria.

(2) A thorough analysis of the accuracy-fairness trade-off between three classifier families (Random Forest, Decision Tree, and Logistic Regression) showed that the cost of mitigation varies depending on the model architecture.

(3) Empirical evidence suggests in-processing techniques are best suited for environments where equality is the main goal, whereas pre-processing solutions produce consistent, affordable fairness benefits, making them a good default for deployments with limited resources.

(4) A practitioner-focused paradigm for making decisions based on observable trade-off patterns that offers recommendations for technique selection based on deployment limitations, legal requirements, and acceptable performance margins.

II. Literature Review

A fundamental tension emerged in the formal study of fairness: the intuitive urge to treat individuals equally does not always translate into a single, mathematically consistent concept. Barocas, Hardt, and Narayanan [1] distinguished between group fairness, which requires statistical parity across demographic categories, and individual fairness, which requires that similar individuals receive similar treatment. This conceptual framework served as the foundation for much later work. The policies that emerged from their concept, such as Equal Opportunity and Demographic Parity Difference Differences are now common tools for assessing classifier equity, and this study uses both.

The focus subsequently shifted to remedial methods. Calders, Kamiran, and Pechenizkiy [2] showed that instance-level weights calibrated to the joint distribution of sensitive attributes and outcome labels could significantly reduce discriminatory patterns in Naïve Bayes classifiers without changing the underlying data structure. This finding made reweighing one of the most practical tools in the fairness toolkit. Later, Kamiran and Calders [3] expanded this line of inquiry to resampling techniques, demonstrating that selectively oversampling and under sampling based on group-outcome combinations enhanced fairness on real income-prediction benchmarks, establishing data-level adjustments as a reliable first line of defense against inherited bias.

Much more sophisticated gear was presented by the in-processing literature. By adding covariance-based fairness restrictions to the logistic regression objective, Zafar et al. [4] turned what had previously been a post-hoc concern into a first-class optimization target. Their method showed that accuracy and fairness do not always have to be viewed as mutually exclusive goals—in some problem geometries, model-level constraints only slightly reduce performance. A different approach was taken by Agarwal et al. [5], who cast fair classification as a constrained optimization problem that could be solved by a series of standard weighted reductions. As a result, practitioners who were already acquainted with off-the-shelf learning libraries could now use in-processing techniques. The other in-processing technique discussed here, adversarial debiasing, involves the addition of an auxiliary classifier that makes an effort to predict sensitive attributes from the representation of the primary model. If the adversary is successful, the main model is penalized, thereby discouraging the encoding of group-identifiable patterns [4].

In situations where retraining is costly or the underlying model is a proprietary black box, post-processing corrections—which were methodically introduced by Hardt, Price, and Srebro [6]—operate solely on predicted scores rather than the model itself. By identifying group-specific decision thresholds that minimize error-rate parity violations, their equalized odds post-processor achieves fairness without affecting training data or model parameters. By adding score calibration to the threshold adjustment, Pleiss et al. [7] improved this line of work by guaranteeing that predicted probabilities maintain their probabilistic interpretation across demographic groups—a feature whose significance had been somewhat overlooked in earlier threshold-based approaches.

The research has seldom directly compared the three intervention families in controlled experimental settings, despite this wealth of individual contributions. Through the AI Fairness 360 toolbox, Bellamy et al. [8] provided a comprehensive assessment; nevertheless, their evaluation included a smaller range of classifiers and lacked the cross-dataset consistency that would enable confident generalizations. In order to close that gap, the current study applies a consistent experimental approach to six methods, two datasets, and three classifiers—conditions that allow the observed trade-off patterns to be directly compared across techniques.

III. Methodology

A. Dataset Selection

Two benchmark datasets were selected because they are indicative of real-world classification contexts where bias has observable societal repercussions, well-understood in the fairness field, and publically accessible for reproducibility.

48,842 records from the 1994 U.S. Census make up the Adult Income dataset, which is obtained from the UCI Machine Learning Repository. Fourteen economic and demographic characteristics are included in each record; the binary result shows if the yearly income is more than $50,000. Because the dataset shows significant variance along this dimension—a mirror of verified pay inequality in the underlying population—gender was chosen as the sensitive feature. This makes it a trustworthy stress test for fairness interventions.

7,214 defendant records from Broward County, Florida, are included in the COMPAS dataset, which ProPublica made public after conducting an investigation in 2016. A binary indicator of re-arrest within two years of release serves as the target variable. In line with the extensively reported racial differences in the COMPAS risk scores that initially prompted criticism of algorithmic criminal risk assessment, race was identified as the sensitive attribute. Incorporating both datasets guarantees that results are not limited to a particular domain, kind of sensitive attribute, or degree of baseline discrepancy.

B. Baseline Models

Selecting Random Forest, Decision Tree, and Logistic Regression establishes a baseline for comparison. One important factor is that it covers a variety of approaches, including crowd-averaged trees, branching rules, and straight-line forecasts. This range makes it possible to determine if remedies for unfairness apply to all sorts of blueprints or just some of them. Does the model's shape affect how effectively changes adhere? The arrangement seeks to ascertain without presuming solutions in advance. Every technique has an own flavor—some transparent, some tangled—but they are all used for testing. Depending on the structure, fairness adjustments may react differently; this combination aids in identifying trends. Diversity here exposes more than repetition ever could since different algorithms respond differently to corrections. Not only are results important, but it is also important to understand where interventions succeed and fail.

C. Bias Mitigation Techniques

examining six intervention strategies that are divided into three stages of the procedure:

Pre-processing - Reweighing [3] gives each training instance a scalar weight determined by combining its result label and sensitive attribute value. In order to change the effective distribution that the classifier sees without changing the actual data values, examples from underrepresented groups with positive labels are upweighted and overrepresented combinations are downweighted. An alternative strategy is used by Disparate Impact Remover [3], which loosens the implicit relationship between feature values and group membership by changing the distributions of non-sensitive features within each group to make them more statistically similar. By controlling the degree of transformation, the repair parameter enables practitioners to weigh the benefits of fairness against the possibility of information loss.

In-processing - An adversarial network whose only job it is to predict the sensitive attribute from the intermediate representations of the primary network is trained alongside a primary classification network using adversarial debiasing. Training pressure builds up against encoding group-distinguishing signals, even those conveyed by seemingly neutral characteristics, because the primary model's loss function penalizes successful adversarial inference. By adding a regularization term to the standard logistic loss that penalizes the mutual information between predictions and sensitive attributes, Prejudice Remover [4] accomplishes a similar goal by deterring the model from taking advantage of group-correlated patterns regardless of their predictive value for the intended label.

Post-processing - By solving a linear program that minimizes violations of the equalized odds criterion—that is, the requirement that both true positive rates and false positive rates be equalized across groups—Equalized Odds Post-Processing [6] determines group-specific decision thresholds. This modification is compatible with any black-box classifier since it only affects output scores, leaving the underlying model unaltered. This framework is expanded upon by Calibrated Equalized Odds [7], which additionally enforces score calibration: the adjusted predictions must maintain consistent probabilistic meaning across groups, guaranteeing that a predicted probability of 0.7 carries the same empirical frequency of positive outcomes regardless of the subject's membership in the advantaged or disadvantaged group.

D. Evaluation Metrics

The accuracy and macro-averaged F1-score were used to evaluate the model's performance. Because Macro-F1 weights each class equally rather than aggregating across instances, it is more revealing under the situation of class imbalance, which is present in both datasets. Two complementing metrics were used to quantify fairness. The absolute difference in positive prediction rates between advantaged and disadvantaged groups is measured by the Demographic Parity Difference (DPD); a DPD of zero denotes that, independent of actual underlying rates, both groups obtain favorable outcomes at equal rates. Equal Opportunity Difference (EOD) measures whether the system is equally effective at detecting positive outcomes for members of each group by limiting attention to qualified individuals and assessing the difference in genuine positive rates across groups. When combined, these measures offer a fair representation of equity that is both outcome-based and opportunity-based.

E. Experimental Setup

Python 3.10 was used to implement each experiment. The baseline classifiers were from Scikit-learn, and the bias reduction methods were taken from IBM's AI Fairness 360 toolset [8]. To maintain class proportions across splits, stratified sampling was used to divide the data into training and test sets in an 80:20 ratio. In order to avoid any leaking of test-set knowledge into model selection, hyperparameters were chosen using five-fold cross-validation that was only applied to the training partition. To guarantee complete consistency, fixed random seeds were used for each trial.

IV. Results and Discussion

A. Baseline Performance

Table I shows accuracy, F1-score, DPD, and EOD across baseline classifiers based on Adult Income dataset findings without any changes. The best model, a Random Forest, performs well, but it exhibits a noticeable imbalance. Thanks in significant part to its adept handling of complex variable connections, its accuracy reaches 86.2% and its F1-score is 0.831. However, there is a paradox here: with a DPD of 0.197, it also produces the worst disparity. Therefore, high prediction quality does not guarantee equitable results for all groups.

Table I. Baseline Model Performance – Adult Income Dataset

Model	Acc. (%)	F1	DPD	EOD
Log. Reg.	82.4	0.798	0.149	0.132
Dec. Tree	83.7	0.811	0.163	0.147
Ran. Forest	86.2	0.831	0.197	0.178

B. Impact of Bias Mitigation

Results for each of the six mitigation strategies used with Logistic Regression on the Adult Income dataset are shown in Table II. Each method lowers both DPD and EOD in comparison to the untreated baseline, demonstrating that the interventions are working as planned. However, there are significant differences in the methods used to accomplish this decrease and the associated costs.

Reweighing and DI Remover, two pre-processing techniques, lower DPD to 0.092 and 0.078, respectively, at the expense of fewer than two percentage points of accuracy. These strategies modify the weighting or marginal distribution of training instances without changing the classifier's learning dynamics, which is consistent with the small accuracy penalty. The most drastic bias reductions are achieved by in-processing techniques (Adversarial Debiasing and Prejudice Remover); DPD drops to 0.043 and 0.051, respectively, but at the expense of 3.1 and 3.4 accuracy points. This trade-off illustrates the clear conflict between penalizing group-correlated learning signals and optimizing for a discriminative objective; the model is being asked to do less with the information at its disposal. Post-processing techniques fall somewhere in the

middle: by maintaining score interpretability across groups, Calibrated Equalized Odds (DPD = 0.071, accuracy = 81.2%) maintains slightly more predictive value than its uncalibrated counterpart (DPD = 0.067, accuracy = 80.7%), indicating that the calibration constraint effectively regularizes the threshold adjustment.

Table II. Post-Mitigation Results – Logistic Regression (Adult)

Technique	Accuracy (%)	F1-Score	DPD	EOD	Technique	Accuracy (%)	F1-Score
Reweighing	81.9	0.787	0.092	0.083	Reweighing	81.9	0.787
DI Remover	80.3	0.774	0.078	0.071	DI Remover	80.3	0.774
Adversarial Debiasing	79.1	0.762	0.043	0.039	Adversarial Debiasing	79.1	0.762
Prejudice Remover	78.8	0.759	0.051	0.046	Prejudice Remover	78.8	0.759
Eq. Odds Post-Processing	80.7	0.776	0.067	0.062	Eq. Odds Post-Processing	80.7	0.776
Calibrated Eq. Odds	81.2	0.781	0.071	0.065	Calibrated Eq. Odds	81.2	0.781

C. Fairness–Accuracy Trade-off Analysis

Despite advances in fairness, overall classifier performance, across all classification types and evaluated datasets, continues to decline. The best method for reducing bias in a classifier is through adversarial debiasing. Adversarial debiasing reduces DPD and EOD more than other methods; however, it does come at a significant cost as classification accuracy drops by three to five percentage points. On the other hand, direct data modification achieves a more accurate balance than adversarial debiasing, as it reduces DPD by approximately four of ten classification cases while having less than two full percentage point loss in classification accuracy. Therefore, when the importance of maintaining forecasting accuracy is critical, direct data modification frequently becomes the best choice.

A consistent trend across the COMPAS data set is that, regardless of model, DPD values remain consistently high and greater than 0.25. This finding indicates that there are substantial problems with the original COMPAS scoring algorithm. The DPD disparities measured in this study exist due to the existence of generally recognized biases that existed prior to the introduction of algorithms. In addition, both datasets exhibit similar responses to adjustments, indicating that these disparities are not random occurrences, but rather predictable responses shaped by the type of adjustment applied.

V. Conclusion

This study aimed to address a practical question: which family of bias mitigation approaches should a practitioner start with given a specified deployment environment and a determined accuracy budget? A formal foundation for that decision is provided by the findings, which are derived from six approaches assessed across two datasets and three classifiers under consistent experimental conditions.

The easiest way to start is with pre-processing methods. In production settings where retraining is costly and model modifications entail compliance overhead, their small accuracy penalties, seamless integration into normal training pipelines, and lack of need to modify the underlying model architecture are important advantages. Reweighing in particular provides a dependable first step for organizations starting their fairness journey or working under strict computational restrictions. In contrast, in-processing techniques function best in environments where fairness is a key design goal and when regulatory frameworks allow for or even require slight accuracy decreases. There are actual costs associated with their strong bias reductions, but those costs are quantifiable and predictable. Post-processing methods are particularly useful for retrofitting fairness into current systems without interfering with operating pipelines since they do not require retraining and may be applied to models that have already been deployed.

These results have the wider consequence that algorithmic fairness is not a single issue with a single fix. Technique selection should be guided by the restrictions imposed by various deployment circumstances, including those related to accuracy, interpretability, computing budget, and the nature of the sensitive feature. The type of comparative data generated here is becoming more and more important because to the rising alignment between empirical fairness research and legal frameworks like the EU AI Act, which specifically mandates bias evaluation for high-risk AI applications. Future research should expand this framework in a number of ways, such as analyzing intersectional fairness across several concurrently sensitive attributes, evaluating the efficacy of mitigation on deep learning architectures and large language model fine-tuning pipelines, and creating ethical standards for acceptable accuracy-fairness trade-offs that can be integrated into workflows for model approval.

In the end, prejudice in the world that produced the data is reflected in bias in machine learning. While important, algorithmic fixes are insufficient. Technical rigor, such as that shown here, and a persistent dedication to the institutional and social conditions that dictate what training data is gathered, how labels are assigned, and whose experiences are fairly represented are necessary for the most resilient route to equitable automated decision-making. The technical aspect of that equation is addressed in this study. The remainder is still a human obligation.

Acknowledgment

It is important to acknowledge the unwavering assistance provided by mentors in the Computer Science and Engineering department at Chitkara University in Punjab; their perspectives influenced every phase of our project. Computation-enabling tools, particularly access to the IBM AI Fairness 360 package, were crucial in testing concepts and received quiet praise.

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